Task 3: Work with YourTTS – Report

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> Fill in the [brackets] with your own results, then delete this line.

1. Introduction

The aim of this work was to test the pretrained YourTTS model on Lithuanian, a language it was not trained on, and to prepare a small Lithuanian dataset for adapting the model. YourTTS can copy a new speaker's voice from a short recording without extra training (zero-shot synthesis). I used Python 3.10 with the Coqui TTS library, librosa and matplotlib.

2. Supported languages

The command `print(tts.languages)` returned `['en', 'fr-fr', 'pt-br']`, which means English, French and Brazilian Portuguese. **Lithuanian is not supported.**

3. English synthesis

I recorded the sentence "This is a test of the speech synthesis system." (`reference_en.wav`). YourTTS then synthesized the same sentence in my voice (`output_en.wav`).

Reference recording: [ ] s long, [ ] Hz, WAV, [16]-bit, [mono].

Quality. [2–3 sentences in your own words: Is it easy to understand? Does it sound like you? Does it sound natural or robotic?]

[Insert plots_en.png here]

Figure 1. English: natural speech (left) and synthesized speech (right).

	Natural	Synthesized
Duration (s)	[ ]	[ ]
Sampling frequency (Hz)	[ ]	16000
Mean F0 (Hz)	[ ]	[ ]

Comparison. The synthesized sentence is [shorter / longer] than my recording, partly because my recording has silence at the start and end. YourTTS outputs 16 kHz audio, so its spectrum stops at 8 kHz, while my recording goes up to [ ] kHz; this makes the synthesized voice sound a bit duller. The mean F0 is [close to / higher than / lower than] mine, and the F0 contour is [flatter], so the intonation is less lively.

4. Lithuanian synthesis

I recorded "Tai yra šnekos sintezės sistemos testas." (`reference_lt.wav`). Because Lithuanian is not supported, the sentence was synthesized with the English setting, `language="en"` (`output_lt.wav`).

Quality. [2–3 sentences in your own words. Usually the voice still sounds like you, but the words are read with English pronunciation, and letters like "š" and "ė" are skipped or said wrongly, so the sentence is hard to understand.]

[Insert plots_lt.png here]

Figure 2. Lithuanian: natural speech (left) and synthesized speech (right).

	Natural	Synthesized
Duration (s)	[ ]	[ ]
Mean F0 (Hz)	[ ]	[ ]

Comparison. The synthesized sentence is [shorter / longer] than my recording ([ ] s vs [ ] s). As in the English case, the spectrum stops at 8 kHz, so sounds like "š" and "s" lose their high frequencies. The F0 contour follows [an English-like / a flat] pattern, not natural Lithuanian intonation. Overall the Lithuanian result is [worse] than the English one: the voice is copied, but the pronunciation is not.

5. Lithuanian dataset (EMO corpus)

I downloaded one archive part of the EMO corpus, [emo-wavs.zip.009], and selected all recordings of one speaker in one emotion. The transcriptions came from `metadata_of_EMO_dataset.csv`. The files were renamed `0001.wav`, `0002.wav`, … and split randomly into 90 % training and 10 % validation:

```
dataset/
├── wavs/  (0001.wav, 0002.wav, ...)
├── metadata_train.csv
└── metadata_val.csv
```

Each metadata line has the form `0001|transcription|transcription`.

	Value
Speaker	[F3]
Emotion	[Z – neutral]
Recordings	[ ]
Total duration	[ ] min
Sampling frequency / bit depth / channels	[ ] Hz / [16] bit / [1]
Duration min / max / average	[ ] / [ ] / [ ] s
Training / validation	[ ] / [ ]

[Insert duration_histogram.png here]

Figure 3. Recording durations.

Most recordings last between [ ] and [ ] s.

6. Conclusions

- YourTTS supports English, French and Brazilian Portuguese, but not Lithuanian.
- English zero-shot synthesis worked [well]: the speech was clear and the voice was [similar] to mine, but it had a narrower spectrum (up to 8 kHz) and [flatter] intonation.
- With Lithuanian the voice was copied, but the pronunciation was [poor], so the model needs fine-tuning on Lithuanian data.
- For this, a dataset of [ ] recordings ([ ] min) from speaker [F3], emotion [Z], was prepared from the EMO corpus.